

ps4_mallick

September 22, 2025

0.0.1 Problem Set 4

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[4]: """
MLE vs OLS estimation of log-wage model
"""

import numpy as np
import pandas as pd
import statsmodels.api as sm
from scipy.optimize import minimize

# Load data
df = pd.read_stata("PSID_data.dta")

# Create hourly wage = income / hours
df["hr_wage"] = df["hlabinc"] / df["hannhrs"]

# Filter conditions
df = df[
    (df["hsex"] == 1) &          # male
    (df["age"].between(25, 60)) & # age 25-60
    (df["hr_wage"] > 7)          # hourly wage > $7
].copy()

# Add log wage
df["ln_wage"] = np.log(df["hr_wage"])

print(df.head())
```

	id68	year	intid	relhh	hannhrs	wannhrs	hlabinc	wlabinc	\
11161	402	1971	1	Head	1523.0	0.0	62928.707031	NaN	
11164	461	1971	4	Head	2010.0	0.0	22660.970703	NaN	
11166	1126	1971	8	Head	2860.0	0.0	29337.865234	NaN	
11173	284	1971	20	Head	2400.0	0.0	76885.437500	NaN	
11175	50	1971	29	Head	3164.0	2000.0	31968.156250	28326.214844	

	nochild	wrace	...	wsex	age	wage	hpersno	wpersno	hyrsed	\
11161	0	1.0	...	2.0	51.0	48.0	1.0	2.0	12.0	

11164	0	1.0	...	2.0	55.0	54.0	1.0	2.0	5.0
11166	1	NaN	...	2.0	25.0	24.0	1.0	2.0	16.0
11173	2	1.0	...	2.0	39.0	36.0	1.0	2.0	16.0
11175	3	1.0	...	2.0	36.0	31.0	1.0	2.0	12.0

	wyrsed	pce	hr_wage	ln_wage
11161	12.0	0.247121	41.318916	3.721320
11164	5.0	0.247121	11.274115	2.422509
11166	12.0	0.247121	10.257995	2.328057
11173	12.0	0.247121	32.035599	3.466848
11175	12.0	0.247121	10.103716	2.312903

[5 rows x 54 columns]

Regression Model The model we estimate is:

$$\ln(w_{i,t}) = \alpha + \beta_1 Educ_{i,t} + \beta_2 Age_{i,t} + \beta_3 Age_{i,t}^2 + \beta_4 Black_{i,t} + \beta_5 Hispanic_{i,t} + \beta_6 OtherRace_{i,t} + \varepsilon_{i,t}$$

```
[5]: # Define variables
df["educ"] = df["hyrsed"]
df["age2"] = df["age"]**2
df["black"] = (df["hrace"] == 2).astype(int)
df["hispanic"] = (df["hrace"] == 5).astype(int)
df["other"] = (~df["hrace"].isin([1, 2, 5])).astype(int)

# Drop missing rows
reg_vars = ["ln_wage", "educ", "age", "age2", "black", "hispanic", "other"]
df = df.replace([np.inf, -np.inf], np.nan)
df = df.dropna(subset=reg_vars)

# Build X and y
X = df[["educ", "age", "age2", "black", "hispanic", "other"]]
X = sm.add_constant(X)
y = df["ln_wage"]

# OLS
ols_model = sm.OLS(y, X).fit()
print(ols_model.summary())

# MLE
init_params = np.append(ols_model.params.values, ols_model.resid.std())
bounds = [(None, None)]*X.shape[1] + [(1e-6, None)]

result = minimize(neg_loglike, init_params, args=(y, X),
                  method="L-BFGS-B", bounds=bounds)
```

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mle_params = result.x
print("MLE coefficients:", mle_params[:-1])
print("MLE sigma:", mle_params[-1])
```

```
c:\ProgramData\anaconda3\Lib\site-
packages\statsmodels\regression\linear_model.py:1966: RuntimeWarning: divide by
zero encountered in scalar divide
    return np.sqrt(eigvals[0]/eigvals[-1])
```

OLS Regression Results

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Dep. Variable:          ln_wage    R-squared:                0.198
Model:                  OLS        Adj. R-squared:            0.198
Method:                 Least Squares    F-statistic:          2824.
Date:                  Mon, 22 Sep 2025    Prob (F-statistic):      0.00
Time:                  20:07:31    Log-Likelihood:         -39958.
No. Observations:      57097    AIC:                    7.993e+04
Df Residuals:          57091    BIC:                    7.998e+04
Df Model:               5
Covariance Type:        nonrobust
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	coef	std err	t	P> t	[0.025	0.975]
const	0.5293	0.038	13.798	0.000	0.454	0.605
educ	0.0764	0.001	90.708	0.000	0.075	0.078
age	0.0603	0.002	31.899	0.000	0.057	0.064
age2	-0.0006	2.29e-05	-24.375	0.000	-0.001	-0.001
black	-0.1627	0.009	-18.251	0.000	-0.180	-0.145
hispanic	0	0	nan	nan	0	0
other	0.0085	0.014	0.623	0.533	-0.018	0.035

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Omnibus:                5941.879    Durbin-Watson:           1.934
Prob(Omnibus):           0.000    Jarque-Bera (JB):        16742.771
Skew:                    0.576    Prob(JB):                 0.00
Kurtosis:                5.390    Cond. No.                 inf
=====
```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The smallest eigenvalue is 0. This might indicate that there are strong multicollinearity problems or that the design matrix is singular.

MLE coefficients: [0.52932931 0.07642395 0.0603013 -0.00055766 -0.16265759 0.

0.00846254]

MLE sigma: 0.48719249536115766

Interpretation of OLS Results

- **Education (0.0764)**: Each additional year of education increases log wages by about **7.6%** on average, holding other factors constant. This is highly significant ($p < 0.000$).
- **Age (0.0603) and Age² (-0.0006)**: Wages rise with age, but at a decreasing rate. The positive age coefficient and negative age² coefficient indicate a concave wage–age profile.
- **Black (-0.1627)**: Being Black is associated with about **16% lower wages** compared to otherwise similar White males, and the result is significant.
- **Hispanic (0.0000)**: The Hispanic dummy shows no effect in this sample. Because the sample doesn't have any hispanic male.
- **Other (0.0085)**: No significant difference in wages for the “Other” race category relative to White.

```
[7]: def neg_loglike(params, y, X):
      """
      Negative log-likelihood for linear regression with normal errors.
      params = [beta coefficients..., sigma]
      """
      beta = params[:-1]
      sigma = params[-1]
      if sigma <= 0:
          return np.inf
      resid = y - X @ beta
      n = len(y)
      ll = -0.5*n*np.log(2*np.pi*sigma**2) - 0.5*np.sum((resid/sigma)**2)
      return -ll
```

```
[8]: years = [1971, 1980, 1990, 2000]
      results = []

      for year in years:
          df_year = df[df["year"] == year].copy()
          df_year = df_year.dropna(subset=["ln_wage"] + reg_vars)

          X = df_year[["educ", "age", "age2", "black", "hispanic", "other"]]
          X = sm.add_constant(X)
          y = df_year["ln_wage"]

          # OLS
          ols_model = sm.OLS(y, X).fit()
          beta1_ols = ols_model.params["educ"]

          # MLE
          init_params = np.append(ols_model.params.values, ols_model.resid.std())
          bounds = [(None, None)]*X.shape[1] + [(1e-6, None)]
          result = minimize(neg_loglike, init_params, args=(y, X),
                           method="L-BFGS-B", bounds=bounds)
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beta1_mle = result.x[X.columns.get_loc("educ")]

results.append({
    "year": year,
    "OLS_beta1": beta1_ols,
    "MLE_beta1": beta1_mle
})

results_df = pd.DataFrame(results)
results_df

```

```

[8]:   year  OLS_beta1  MLE_beta1
0  1971    0.066500    0.066500
1  1980    0.066003    0.066003
2  1990    0.095499    0.095499
3  2000    0.110329    0.110329

```

Interpretation of (Education) The coefficient on education () indicates the **percentage increase in wages** from an additional year of schooling.

- In **1971**, returns were about **6.7%**.
- In **1980**, they remained stable at about **6.6%**.
- By **1990**, returns increased to roughly **9.6%**.
- In **2000**, they rose further to around **11.0%**.

Conclusion: Returns to education were relatively flat in the 1970s–1980s but rose significantly in the 1990s and 2000s. This suggests that education has become increasingly rewarded in the labor market over time.

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